

Chapter 8 - Router Types, Ports & Cables

1. Types of Routers Provided by Cisco

Cisco provides **two main types of routers**  :

◆ Unmanaged Routers

- No configuration changes allowed 
- Mostly used in **homes** 
- Plug and play devices

Daily Life Example:

- Like a basic TV remote  with fixed buttons

◆ Managed Routers

- Used in **companies & enterprises** 
- Fully configurable
- High security & control

Managed routers are of **two types** 

2. Mountable vs Non-Mountable Routers

◆ Mountable Routers

- Ports/modules can be **added or removed** 
- Supports **1, 2, 4 or more extra ports**
- Router must be **powered OFF** before adding interfaces 
- Includes **Console & AUX ports**

🧠 Example:

- Like upgrading RAM in a laptop 🖥️
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◆ Non-Mountable Routers

- Fixed hardware
- Usually **4 LAN ports only**
- Ports **cannot be changed**

🧠 Example:

- Like a sealed smartphone 📱
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🔌 3. Important Router Ports

🖥️ Console Port

- Used to **configure router physically**
- Connects router to **laptop/PC** using console cable
- Requires **PuTTY application**

📌 Purpose:

- Initial configuration
- Password recovery

🧠 Example:

- Like using USB cable to configure a new phone 📱
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🔑 AUX Port

- Used in **older Cisco routers**
- Allowed remote access using modem 📞

- Rarely used in modern routers

Telnet & SSH

Used for **remote router access** 

Protocol Security

Telnet  Not secure

SSH  Secure (Encrypted)

USB Port

- Used when router OS is corrupted 
- Cisco IOS can be **reinstalled using USB**

 Example:

- Like reinstalling Windows using a pen drive 

4. LAN Port Types & Speeds

LAN ports connect **PCs, switches, routers** 

Port Type	Speed	Usage
Ethernet (Eth)	10 Mbps	Very old networks
Fast Ethernet (Fa)	100 Mbps	Old networks
Gigabit Ethernet (Gig)	1 Gbps	Modern networks
10 Gig Ethernet	10 Gbps	Data centers
25G / 40G / 100G	Ultra Fast	ISPs & Cloud

5. RJ45 Ethernet Cable Basics

RJ45 cable has **4 twisted pairs (8 wires)** 

◆ **Straight-Through Cable**

Used to connect **different devices** 

 Example:

- PC ↔ Switch
- Router ↔ Switch

◆ **Wire Order (Both Ends Same – T568B)**

1. White-Green
 2. Green
 3. White-Orange
 4. Blue
 5. White-Blue
 6. Orange
 7. White-Brown
 8. Brown
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◆ **Crossover Cable**

Used to connect **same devices** 

 Example:

- PC ↔ PC
- Switch ↔ Switch

◆ **Wire Order**

Side A (T568B):

1. White-Green
2. Green
3. White-Orange
4. Blue
5. White-Blue
6. Orange
7. White-Brown
8. Brown

Side B (T568A):

1. White-Orange
 2. Orange
 3. White-Green
 4. Blue
 5. White-Blue
 6. Green
 7. White-Brown
 8. Brown
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6. How to Download PuTTY

- 1 Open web browser
 - 2 Search **PuTTY download**
 - 3 Open official website
 - 4 Select Windows version (32-bit or 64-bit)
 - 5 Download .exe file
 - 6 Install and open PuTTY
 - 7 Select **Serial / COM port** for console access
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7. Checking IP Details Using CMD

Command:

ipconfig

Shows:

- Private IP Address
 - Subnet Mask
 - Default Gateway
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8. Cisco Packet Tracer Software

Used for **network simulation & practice**

◆ Why Packet Tracer?

- Practice CCNA labs
 - Design networks virtually
 - No physical devices needed
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How to Download Cisco Packet Tracer

- 1 Open browser 
 - 2 Search **Cisco Packet Tracer download**
 - 3 Login using Cisco NetAcad account
 - 4 Choose correct OS (Windows/Linux)
 - 5 Download installer
 - 6 Install and launch Packet Tracer
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Key Takeaways

- Cisco routers can be managed or unmanaged
 - Console port is used for initial setup
 - SSH is more secure than Telnet
 - Cable type depends on device connection
 - Packet Tracer is essential for CCNA practice 
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◆ 1 Chapter Summary

This chapter focuses on **routers, their types, ports, cables, and practical tools** used in real networking environments.

You learned:

- Types of Cisco routers (Managed vs Unmanaged)
- Difference between **Mountable & Non-Mountable** routers
- Important router ports (Console, AUX, USB, LAN)
- Difference between **Telnet and SSH**
- LAN port speeds (Ethernet → 100G)
- Ethernet cable types (**Straight-through & Crossover**)
- RJ45 wiring standards (T568A & T568B)
- Use of **PuTTY** for router configuration
- Checking IP details using **CMD**
- Importance of **Cisco Packet Tracer** for CCNA practice

👉 This chapter builds **hands-on networking foundation** 🧠💻

◆ 2 Chapter Conclusion 🎯

Routers are the **backbone of any network** 🌐.

Understanding router types, ports, and cables is **mandatory** for:

- Network Engineers 💻
- CCNA / CCNP aspirants 📖
- Data center & enterprise networking 🏢

Key takeaway:

- **Managed routers** = control + security 🔒
- **Console port** = first step of router life 🚦
- **SSH > Telnet** always ✅

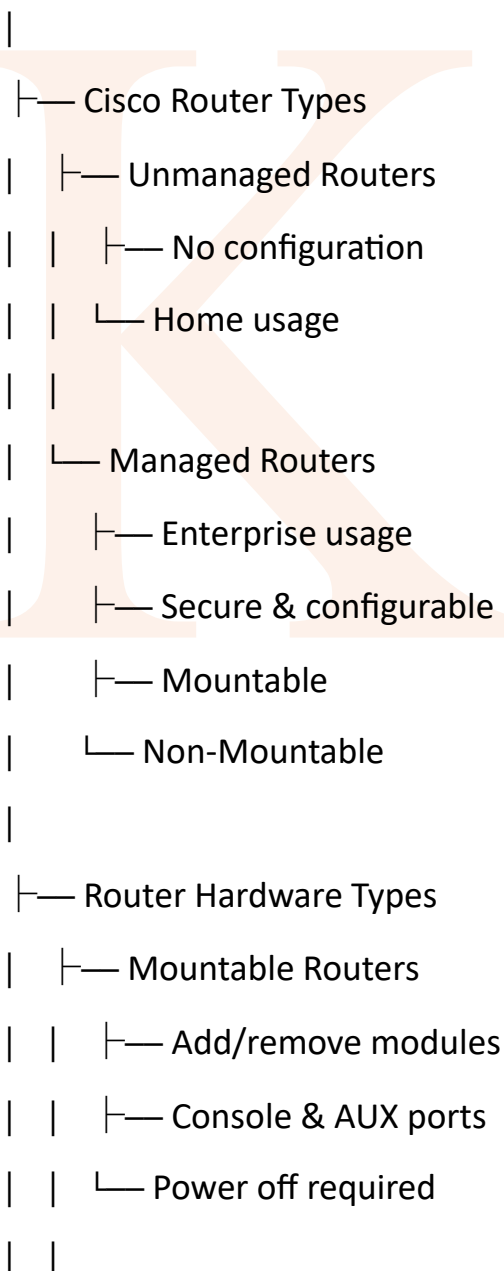
- **Correct cable selection** avoids network failure ❌
- **Packet Tracer** helps you practice without hardware 💡

💬 Interviewer mindset:

“If he knows ports + cables clearly, he can work on real devices.” 🔥

◆ 3 Detailed Mind Map

Lecture 8: Router Types, Ports & Cables



- | └─ Non-Mountable Routers

- | └─ Fixed ports

- | └─ Limited expansion

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- | └─ Router Ports

- | └─ Console Port

- | └─ AUX Port

- | └─ USB Port

- | └─ LAN Ports

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- | └─ Remote Access

- | └─ Telnet (Not secure)

- | └─ SSH (Encrypted)

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- | └─ LAN Port Speeds

- | └─ Ethernet (10 Mbps)

- | └─ Fast Ethernet (100 Mbps)

- | └─ Gigabit (1 Gbps)

- | └─ 10G Ethernet

- | └─ 25G / 40G / 100G

- |

- | └─ Ethernet Cables

- | └─ Straight-Through

- | | └─ Different devices

- | |

- | └─ Crossover

- | └─ Same devices

- |

- |─ RJ45 Standards

- | └─ T568A

- | └─ T568B

- |

- |─ Tools

- | └─ PuTTY

- | └─ Cisco Packet Tracer

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- |─ Commands

- | └─ ipconfig

◆ 4 Q & A

? Q1. What are the types of Cisco routers?

Answer:

Cisco provides **two types of routers**:

- **Unmanaged routers** (no configuration, home use 🏠)
 - **Managed routers** (enterprise use, fully configurable 🏢)
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? Q2. What is an unmanaged router?

Answer:

An unmanaged router is a **plug-and-play device** where configuration changes are **not allowed**.

Used mostly in **home networks**.

? Q3. Why are managed routers used in companies?

Answer:

Because managed routers provide:

- Security 🔒
 - Traffic control 🚦
 - Custom configuration
 - Better monitoring
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? Q4. What is a mountable router?

Answer:

A mountable router allows **hardware modules or ports to be added or removed**.

Router must be **powered OFF** before adding interfaces ⚠️.

? Q5. Difference between mountable and non-mountable routers?

Answer:

Feature	Mountable	Non-Mountable
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Ports	Changeable	Fixed
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Flexibility	High	Low
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Usage	Enterprise	Small networks
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? Q6. What is a console port?

Answer:

A console port is used for **initial router configuration** by physically connecting a PC using a **console cable**.

? Q7. When is console port used?

Answer:

- First-time router setup
 - Password recovery
 - When network access is not available
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? Q8. What software is used with console port?

Answer:

PuTTY is used to access router CLI via console port  .

? Q9. What is an AUX port?

Answer:

AUX port allows **remote access using modem**.

It is mostly **obsolete** now  .

? Q10. Difference between Telnet and SSH?

Answer:

Protocol Security

Telnet ❌ Plain text

SSH ✅ Encrypted

👉 SSH is always preferred 🔒 .

? Q11. What is the use of USB port in a router?

Answer:

USB port is used to **reinstall Cisco IOS** when the router OS is corrupted ⚠️ .

? Q12. What are LAN ports used for?

Answer:

LAN ports connect:

- PCs 🖥️
 - Switches 🔄
 - Routers 🌐
-

? Q13. Explain LAN port speeds.

Answer:

- Ethernet → 10 Mbps
 - Fast Ethernet → 100 Mbps
 - Gigabit → 1 Gbps
 - 10G / 25G / 40G / 100G → Data centers & ISPs 🚀
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? Q14. What is an RJ45 cable?

Answer:

RJ45 is an **8-wire twisted-pair Ethernet cable** used for LAN communication 🔌.

? Q15. What is a straight-through cable?

Answer:

Used to connect **different devices** like:

- PC ↔ Switch
- Router ↔ Switch

Both ends use **same wiring standard (T568B)**.

? Q16. What is a crossover cable?

Answer:

Used to connect **same devices** like:

- PC ↔ PC
- Switch ↔ Switch

Uses **T568A** on one end and **T568B** on the other.

? Q17. What is T568A and T568B?

Answer:

They are **RJ45 wiring standards** that define the color order of Ethernet wires



? Q18. Which cable is mostly used today?

Answer:

Straight-through cable, because modern devices support **auto MDI-X** 🔄.

? Q19. What command is used to check IP details?

Answer:

ipconfig

It shows IP address, subnet mask, and gateway 🖨️.

? Q20. What is Cisco Packet Tracer?

Answer:

Cisco Packet Tracer is a **network simulation tool** used to practice CCNA labs without real hardware 🧠💻.

? Q21. Why is Packet Tracer important for students?

Answer:

Because it allows:

- Virtual network design
 - Command practice
 - Real-world scenarios
 - Zero hardware cost 💰
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? Q22. Real-life example of router?

Answer:

A router works like a **traffic police** 🚦, deciding which data goes where.
